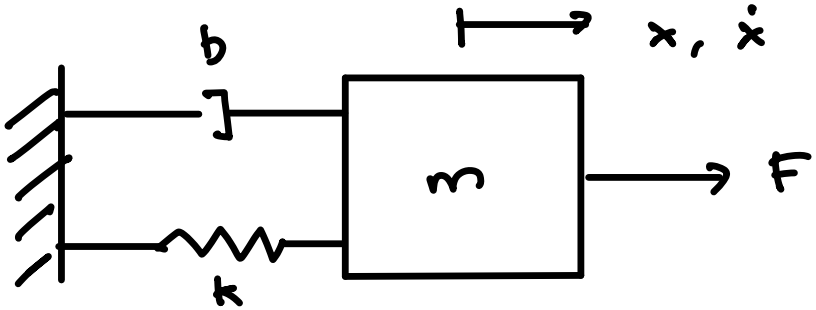
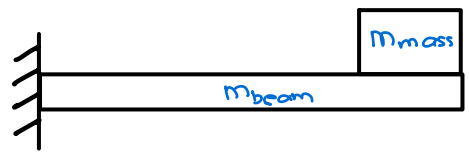



$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = F$$

Assume $x(0), \dot{x}(0) = 0$

$$m\ddot{x} + b\dot{x} + kx = F$$

$$\ddot{x} + \frac{b}{m}\dot{x} + \frac{k}{m}x = \frac{F}{m}$$

$$\frac{X(s)}{U(s)} = \frac{1/m}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = u$$

where $\omega_n = \sqrt{k/m}$, $\zeta = \dfrac{b}{2\sqrt{km}}$, $u = F/m$

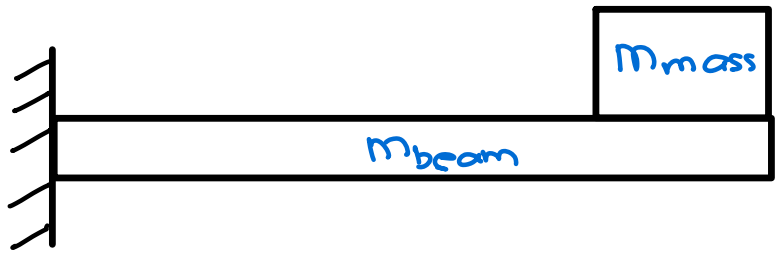
natural freq

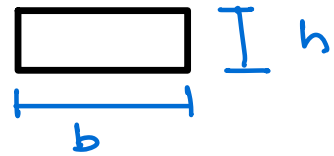
damping ratio

Parameters to find: $\omega_n$, $\zeta$

Natural freq, $\omega_n = \sqrt{k/m_{eff}}$ $\qquad T_n = 2\pi/\omega_n$

Damped freq, $\omega_d = \omega_n\sqrt{1-\xi^2}$ $\qquad T_d = 2\pi/\omega_d$




Finding $\omega_n$,

$$k = \frac{3EI}{l^3}$$

$$m_{eff} = m_{mass} + 0.23\, m_{beam}$$

E → young's modulus

$I = \frac{1}{12} bh^3$ for rect. beam

$l$ → length of beam

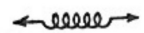
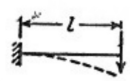
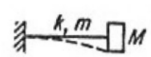
APPENDIX

J.P. Den Hartog
Mechanical Vibrations, Dover Publ., New York, 1934.

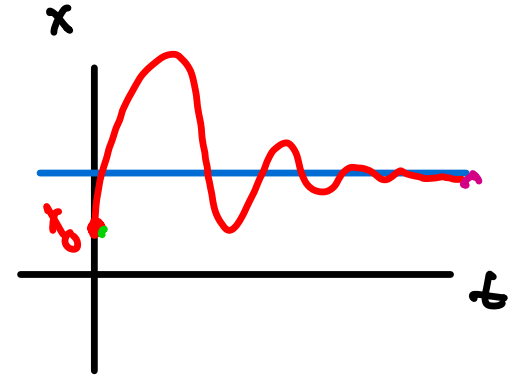
A COLLECTION OF FORMULAS

I. Linear Spring Constants

("Load" per inch deflection)

| | | | |
|---|---|---|---|
|  | Coil dia. $D$; wire dia. $d$; $n$ turns | $k = \frac{Gd^4}{8nD^3}$ | (1) |
|  | Cantilever | $k = \frac{3EI}{l^3}$ | (2) |
|  | Cantilever; end mass $M$; beam mass $m$, stiffness by formula (2) | $\omega_n = \sqrt{\frac{k}{M + 0.23m}}$ | |

# Finding $\xi$ through log decrement



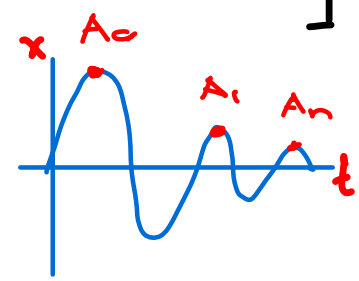

For underdamped 2nd order response

$$\ddot{x} + 2\xi\omega_n\dot{x} + \omega_n^2 x = 0$$

↳ unforced $F(t) = 0$

soln: $$x(t) = e^{-\xi\omega_n t}\left[\frac{\dot{x}_0 + \xi\omega_n x_0}{\omega_d}\sin(\omega_d t) + x_0\cos(\omega_d t)\right]$$

$$x(t) = \underbrace{A_0}_{\text{amplitude}}\ \underbrace{e^{-\xi\omega_n t}}_{\text{decay}}\ \underbrace{\cos\left[\omega_d t - \phi\right]}_{\text{sinusoidal behavior}}$$



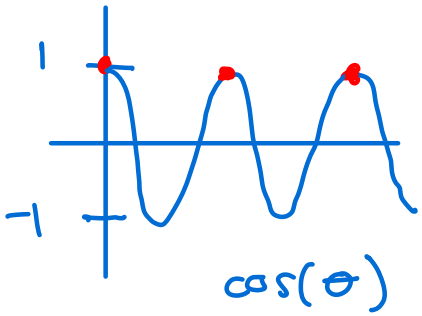

Amplitude, $$A_0 = \left[\left(\frac{\dot{x}_0 + \xi\omega_n x_0}{\omega_d}\right)^2 + x_0^2\right]^{1/2}$$

$$\tan\phi = \frac{\dot{x}_0 + \xi\omega_n x_0}{\omega_d x_0}$$

since we know the decay has a relationship with $\xi$ damping, we can experimentally get amplitude ratio $\frac{A_n}{A_{n+1}}$ to find damping ratio $\xi$

$$\text{If } x(t) = A_n = A_0 e^{-\xi\omega_n t_n} \underline{\cos[\omega_d t_n - \phi]}$$

$= +1$ at positive peaks




$$\frac{A_n}{A_{n+1}} = \frac{\cancel{A_0} e^{-\xi\omega_n t_n}}{\cancel{A_0} e^{-\xi\omega_n t_{n+1}}}$$

$$\ln\left[\frac{A_n}{A_{n+1}} = \exp\left[\xi\omega_n \underline{(t_{n+1} - t_n)}\right]\right]$$

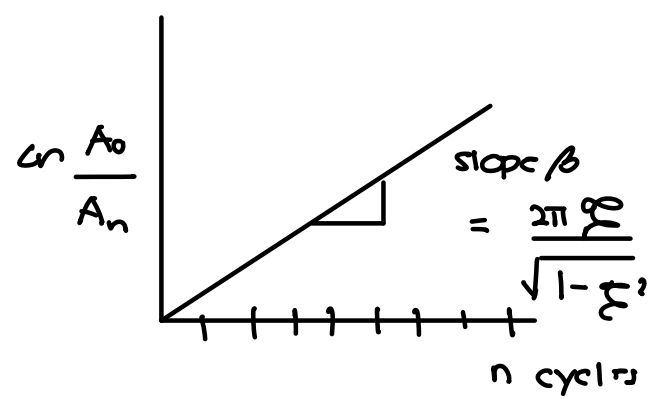
$= T_d$

$$\ln\left[\frac{A_n}{A_{n+1}}\right] = \xi\cancel{\omega_n}\left(\frac{2\pi}{\cancel{\omega_n}\sqrt{1-\xi^2}}\right)$$

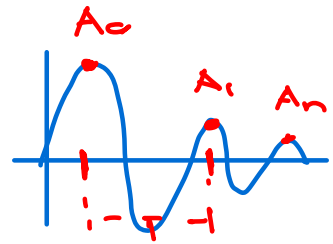
$= 2\pi/\omega_d$

$= \frac{2\pi}{\omega_n\sqrt{1-\xi^2}}$

$$\ln\left[\frac{A_0}{A_n}\right] = \underbrace{\frac{2\pi\xi}{\sqrt{1-\xi^2}}}_{\beta} \cdot n$$

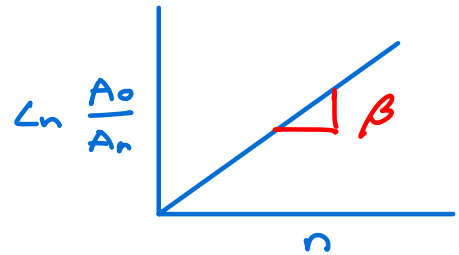


In practice, after data




→ Get peaks $A_n$, period $T$

→ Plot $\ln \frac{A_0}{A_n}$ vs. $n$




→ Get $\beta$

→ solve for $\xi$ damping ⟶ solve for $\omega_d$ if needed

$$\beta = \frac{2\pi\xi}{\sqrt{1-\xi^2}}$$

$$\beta\left(\sqrt{1-\xi^2}\right) = 2\pi\xi$$

$$\beta^2(1-\xi^2) = 4\pi^2\xi^2$$

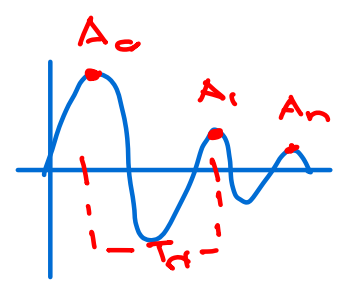
$$\beta^2 = 4\pi^2\xi^2 + \beta^2\xi^2$$

$$\xi = \frac{\beta^2}{\sqrt{4\pi^2+\beta^2}}$$

$$\omega_d = \omega_n\sqrt{1-\xi^2}$$

or

$$T_d = \frac{2\pi}{\omega_d}$$



## verifying effective mass

$$\omega_d = \omega_n \sqrt{1-\zeta^2}$$

$$\omega_n = \frac{\omega_d}{\sqrt{1-\zeta}}$$

$$\omega_n = \sqrt{k/m_{eff}}$$

$$m_{eff} = k/\omega_n^2$$